بسم الله الرحمن الرحيم

«وما أوتيتم من العلم الا قليلا»



Pulmonary Embolism:

BY

Mohammad Gamal Abdalrahman

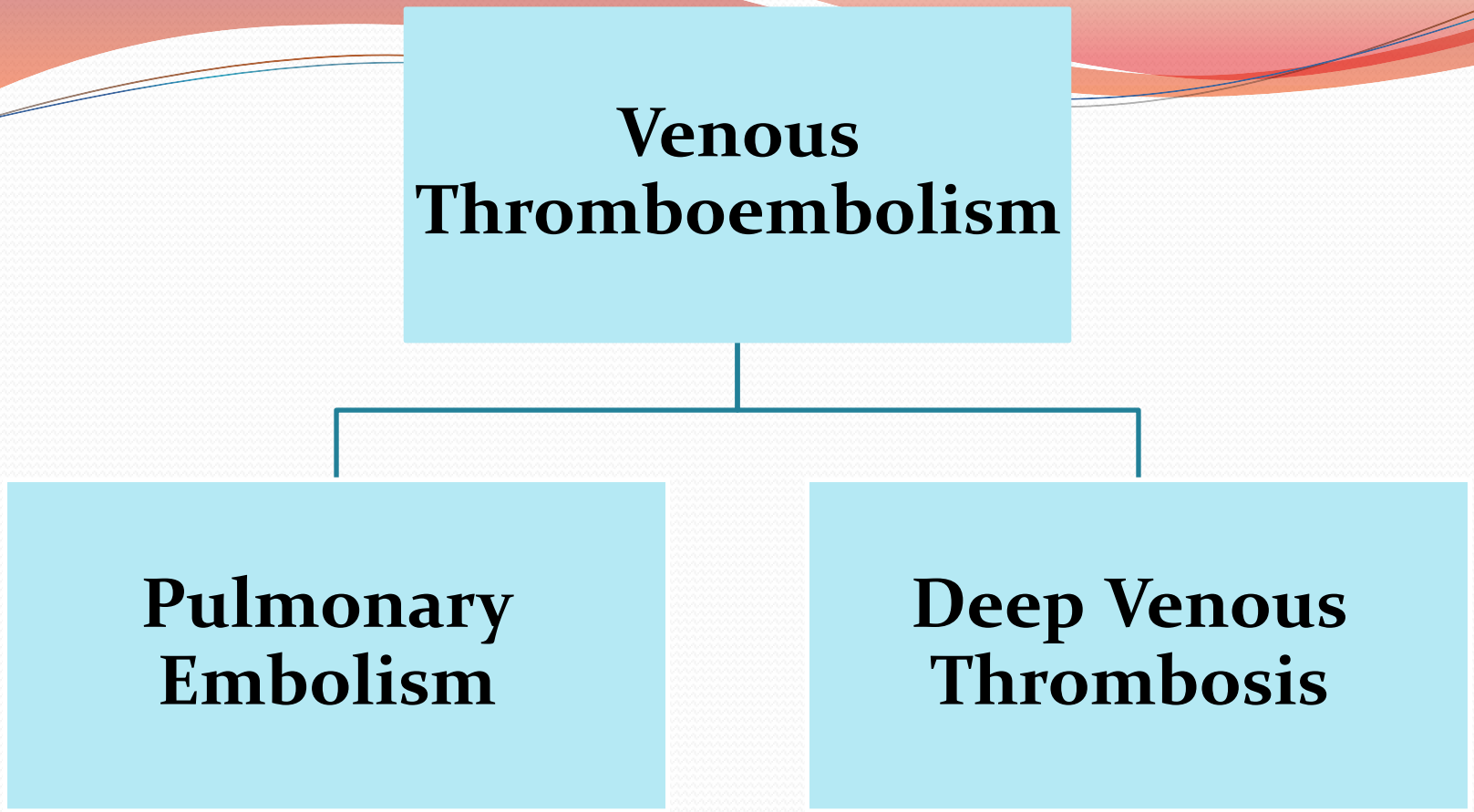
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Introduction

- The word "embolism" comes from the Greek émbolos, meaning "stopper" or "plug."
- Pulmonary embolism (PE) is the blockage of the pulmonary artery or one of its branches by an embolus.
- PE is a relatively common cardiovascular emergency.



Among patients with proximal DVT, 50% have an associated PE at lung scan.

About 70% of patients with PE, have DVT

Prevalence

- Though only 40–53 per 100 000 persons were diagnosed with PE per year, the annual incidence in the United States was estimated at 600 000 cases.
- The true incidence of PE is difficult to assess in view of its non-specific clinical presentation.

Predisposing factors

- Patient-related predisposing factors: (usually permanent)
- Setting-related predisposing factors: (temporary)

In 40% of patients with PE, no predisposing factors are found.

Strong risk factors (OR>10)
Fracture of lower limb
Previous VTE
Spinal cord injury
Hospitalization for heart failure or atrial fibrillation/flutter (within previous 3 months)
Hip or knee replacement
Major trauma
Myocardial infarction (within previous 3 months)

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VTE = venous thromboembolism.

Moderate risk factors (OR 2–9)
Arthroscopic knee surgery
Autoimmune diseases
Blood transfusion
Central venous lines
Intravenous catheters and leads
Chemotherapy
Congestive heart failure or respiratory failure
Erythropoiesis-stimulating agents
Hormone replacement therapy (depends on formulation)

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VTE = venous thromboembolism.

Moderate risk factors (cont'd)
In vitro fertilization
Oral contraceptive therapy
Postpartum period
Infection (specifically pneumonia, urinary tract infection, and HIV)
Inflammatory bowel disease
Cancer (highest risk in metastatic disease)
Paralytic stroke
Superficial vein thrombosis
Thrombophilia

VTE = venous thromboembolism.

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Weak risk factors (OR <2)
Bed rest >3 days
Diabetes mellitus
Arterial hypertension
Immobility due to sitting (e.g. prolonged car or air travel)
Increasing age
Laparoscopic surgery (e.g. cholecystectomy)
Obesity
Pregnancy
Varicose veins

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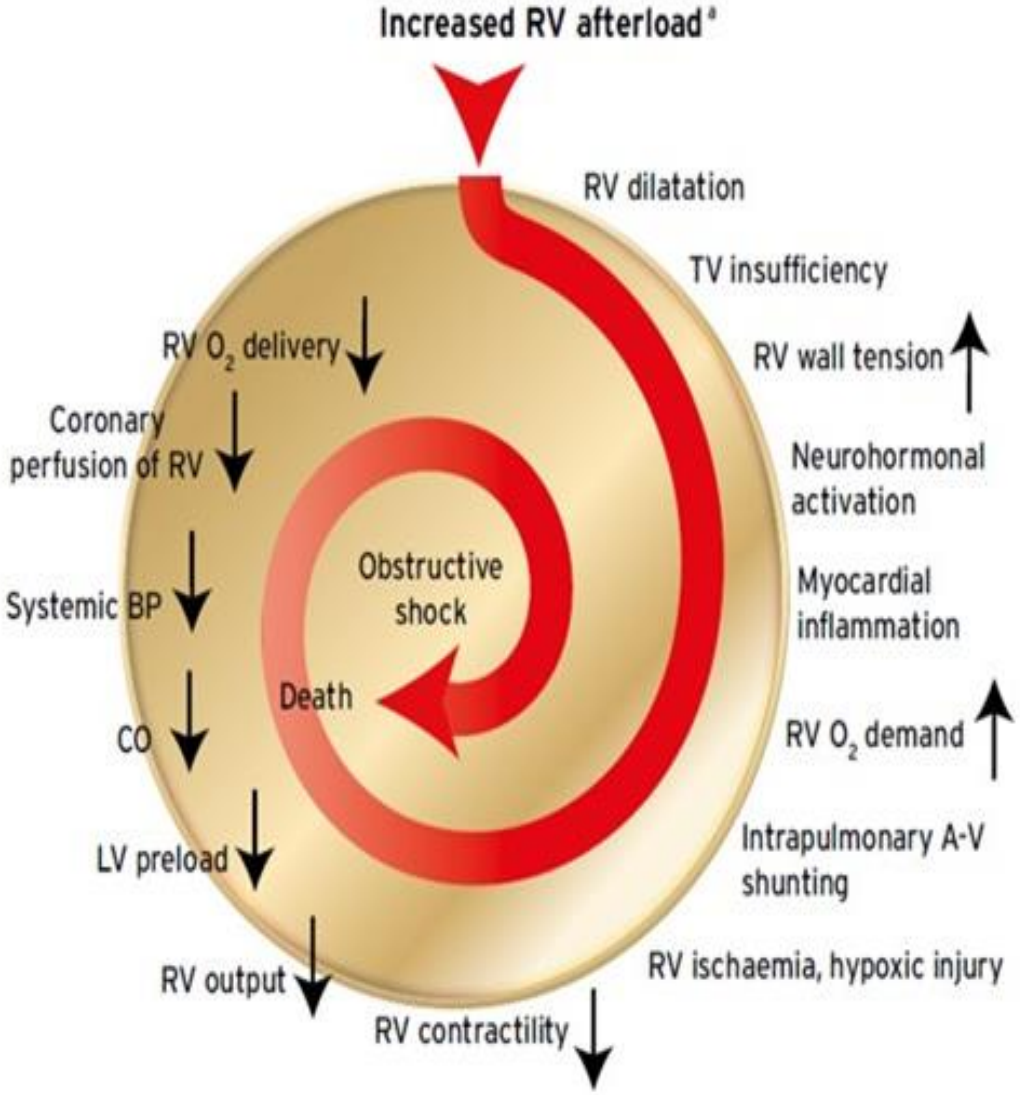
VTE = venous thromboembolism.

Pathophysiology

- **Pulmonary artery pressure (PAP) increases if >30-50% of the total cross-sectional area of the pulmonary arterial bed is occluded by thromboemboli.**

Figure 1 The spiral of haemodynamic collapse in acute PE

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Clinical presentation

Symptoms:

- Dyspnoea (80%)
- Chest pain (pleuritic) (52%)
- Chest pain (substernal) (12%)
- Cough (20%)
- Haemoptysis (11%)
- Syncope (19%)

Signs

- Tachypnoea (>20/min) (70%)
- Tachycardia (>100/min) (26%)
- Signs of DVT (15%)
- Fever (>38.50C) (7%)
- Cyanosis (11%)

Diagnosis

- Symptoms, clinical signs, and routine laboratory tests do not allow the exclusion or confirmation of acute PE but increase the index of its suspicion.

Assessment of clinical probability

- The most frequently used clinical prediction rule is the Canadian rule, by Wells et al. The revised Geneva rule is also used in Europe.

The BTS pre-test clinical probability scoring system:

a) Patient has clinical features compatible with PE:

- raised respiratory rate
- ±haemoptysis
- ±pleuritic chest pain.

Plus 2 other factors:

1- Absence of another reasonable clinical explanation.

2- Presence of a major risk factor.

a) plus 1 and 2: HIGH pre-test clinical probability

a) plus 1 or 2: INTERMEDIATE pre-test clinical probability

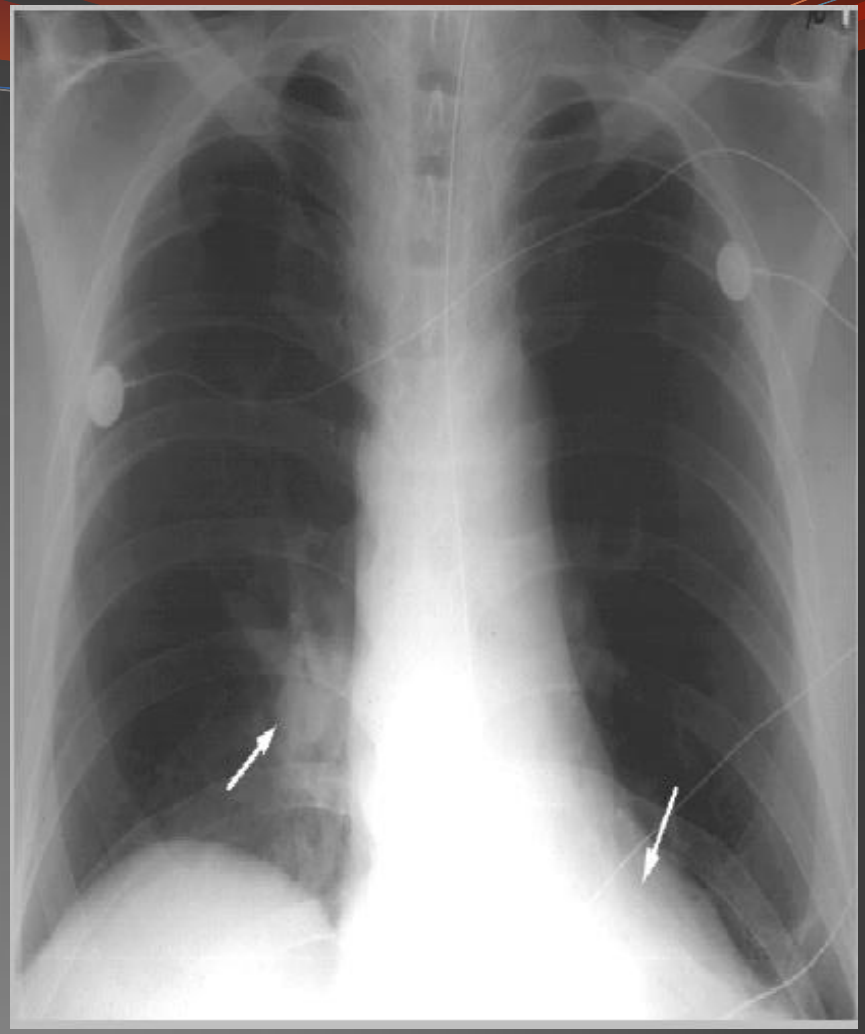
a) alone: LOW pre-test clinical probability.

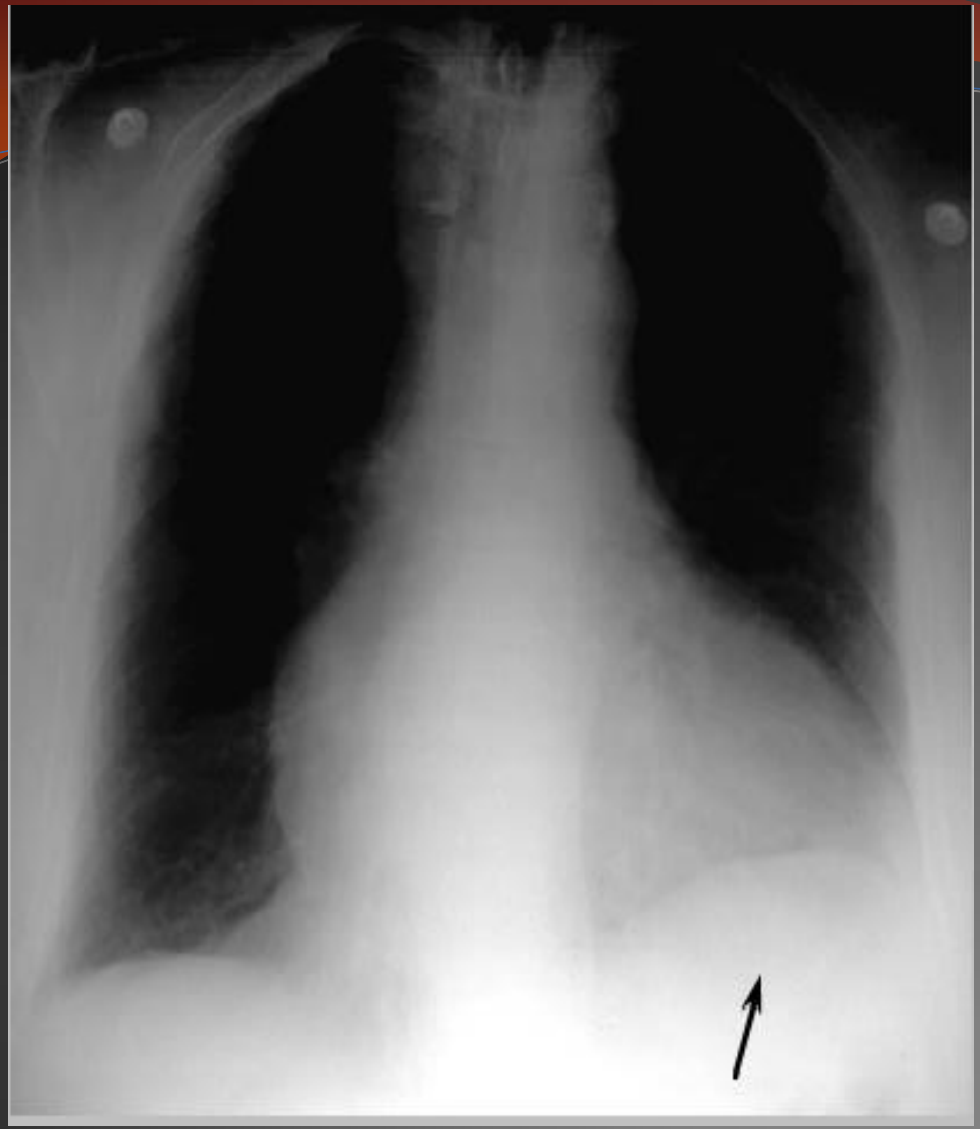
Investigations

1- Plain Chest X-ray: usually abnormal, CXR is very useful in excluding other causes of dyspnoea and chest pain.

- Normal (14%)
- Atelectasis or parenchymal density (68%)
- Pleural Effusion (48%)
- Pleural based opacity (35%)
- Elevated diaphragm (24%)
- Prominent central pulmonary artery (15%)
- Cardiomegally (7%)







Elevated hemidiaphragm



Pleural effusion

2- ABG:

- Hypoxaemia,
- Hypocapia and respiratory Alkalosis
- Up to 20% of patients with PE have a normal PaO2

3-ECG: Electrocardiographic signs of RV strain (Non specific):

- Sinus tachycardia
- Inversion of T waves in leads V1–V4,
- QR pattern in lead V1,
- Classic S1Q3T3 type,
- Incomplete or complete right bundle-branch block.

4- D-dimer:

- Plasma D-dimer is a degradation product of cross-linked fibrin.
- Good negative test.
- Plasma D-dimer measurement is recommended in patients with low or intermediate clinical probability, to reduce the need for imaging and irradiation.

5-Compression U/S and CT venography

In 90% of patients, PE originates from DVT in a lower limb. DVT was found in 70% of patients with proven PE. CUS is of great value in patients with contraindications to contrast dye and/or irradiation.

6- Ventilation–perfusion scintigraphy

- Lung Scintegraphy
- planar ventilation/perfusion scan
- V/Q scan

IV injection of technetium (Tc)-99m labelled macroaggregated albumin particles, which block a small fraction of pulmonary capillaries and thereby enable scintigraphic assessment of lung perfusion at the tissue level.

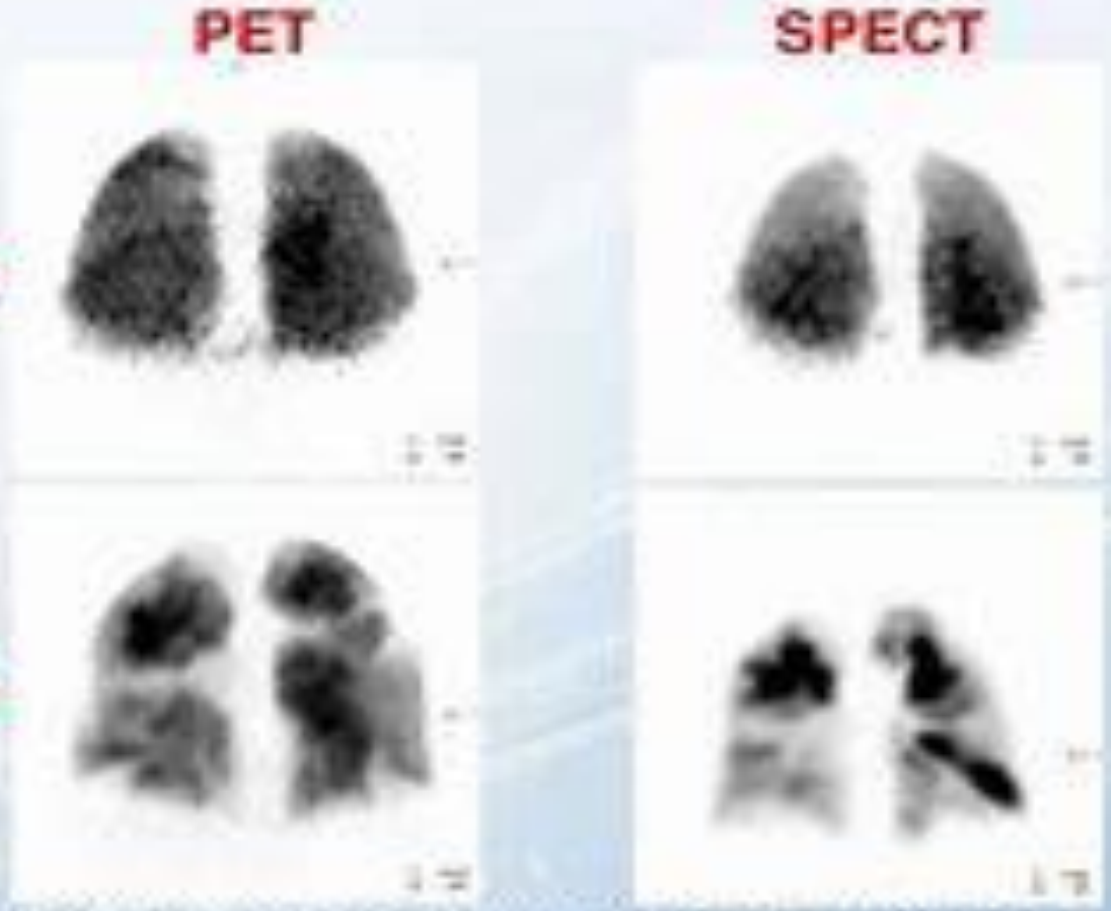
Lung scan results are frequently classified into four categories:

- Normal or near-normal
- low probability
- intermediate probability

→ Non-diagnostic

- high probability of PE >>>>>>>>>>>>>>>> Diagnostic

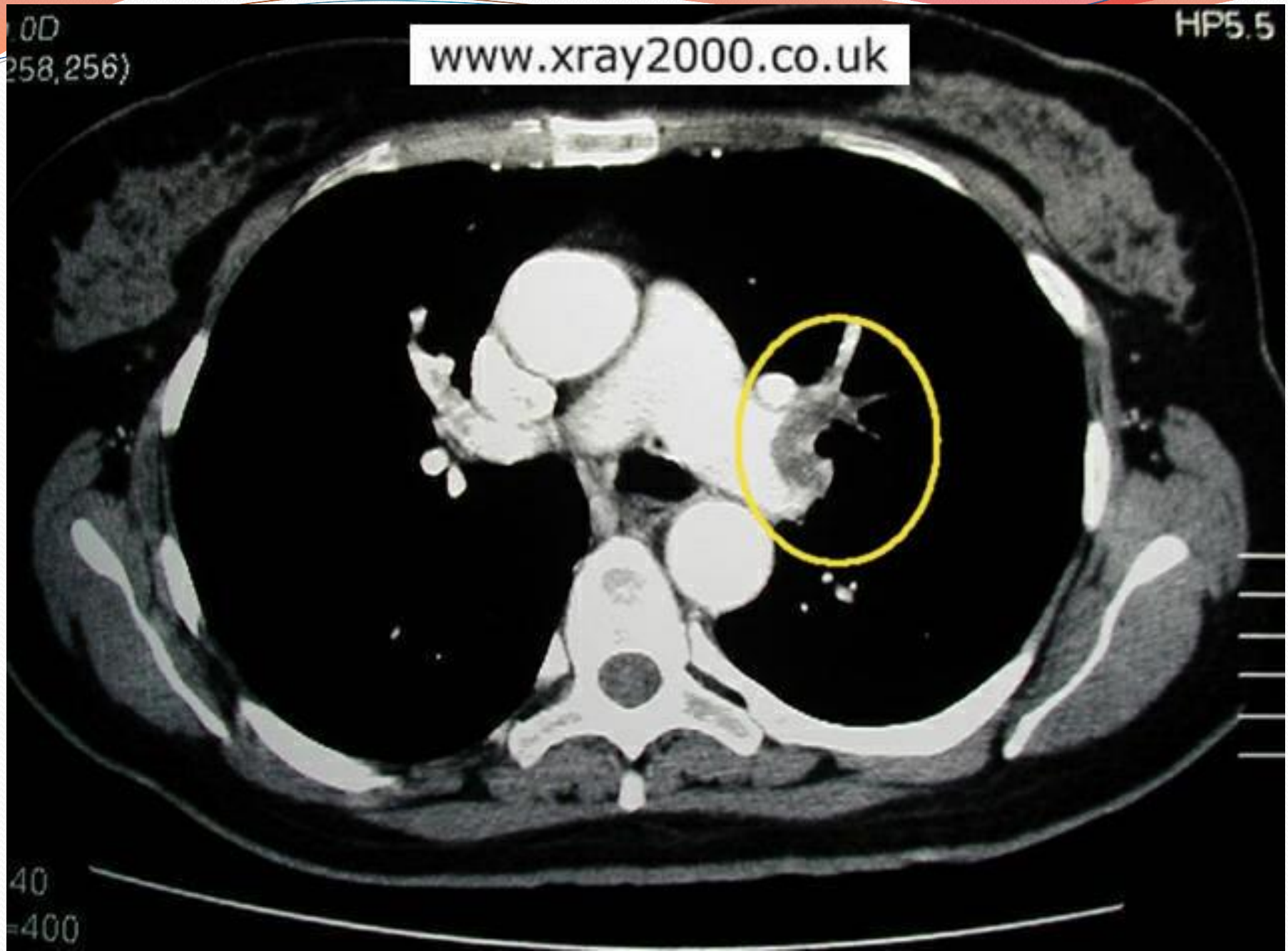
V/Q PET/CT versus V/Q SPECT/CT



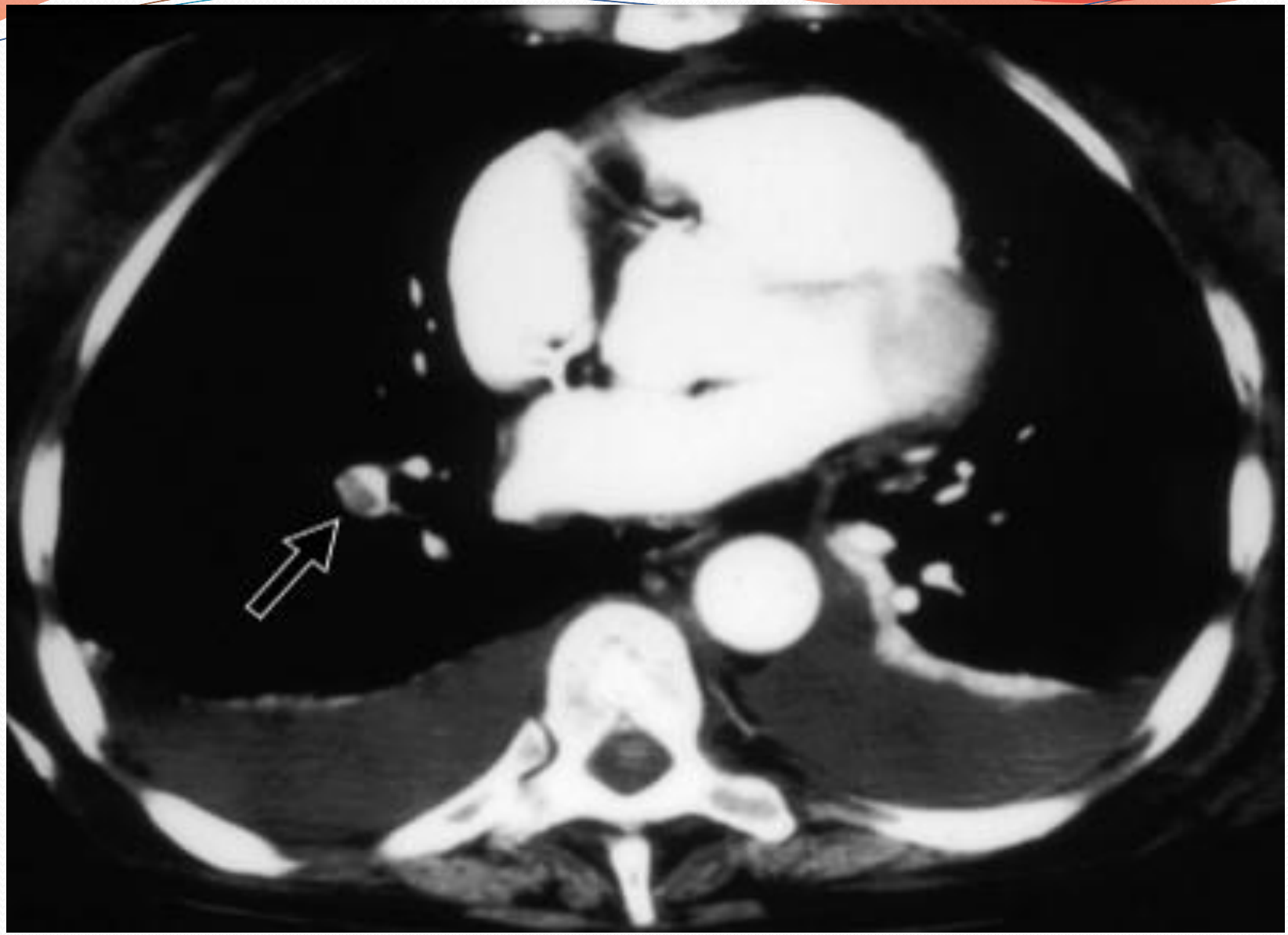
Acute shortness of breath & pleuritic chest pain

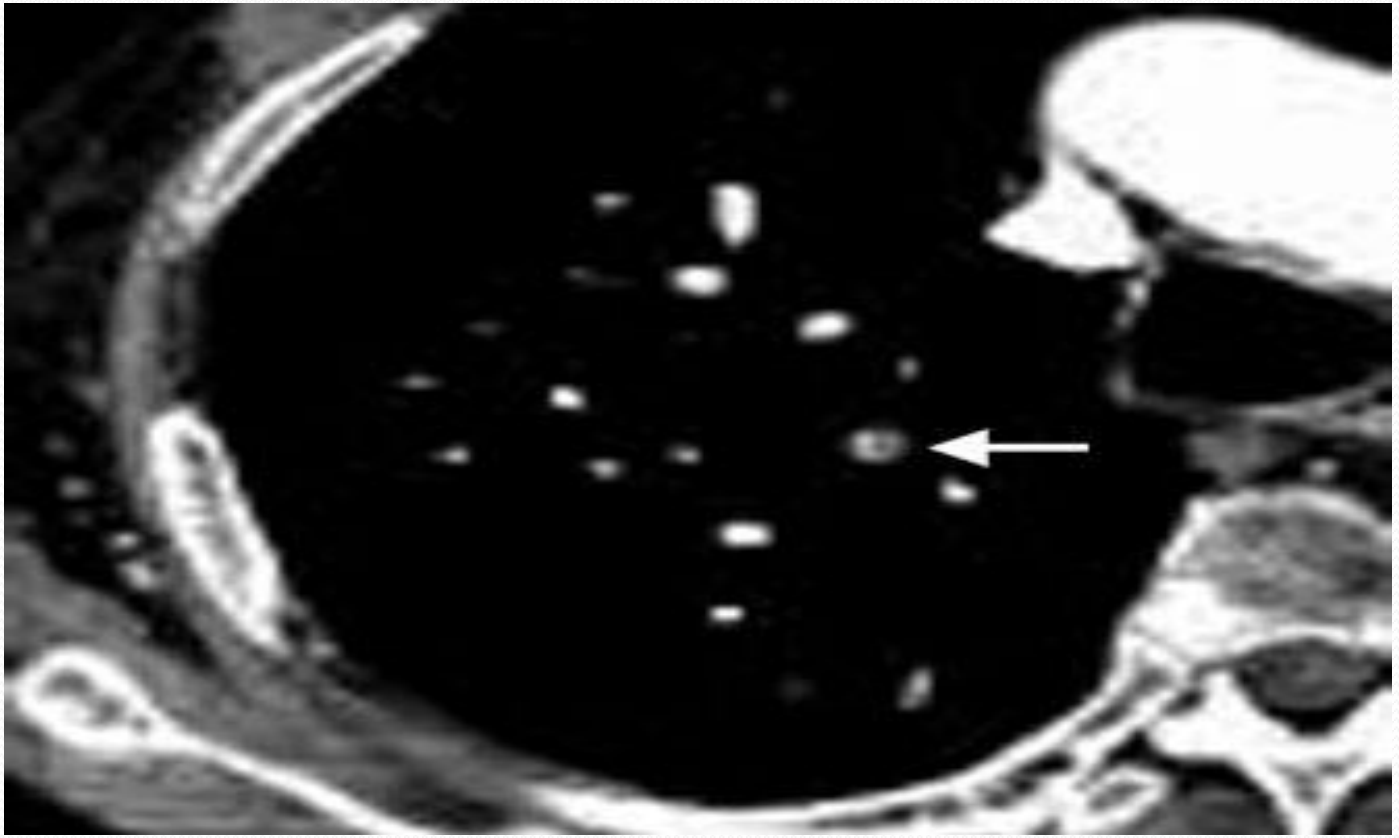
7-Computed tomography

- Gold standard
- CT angiography showing a segmental or more proximal thrombus confirms PE (1B).
- Normal CT angiography may safely exclude PE in patients with high clinical probability (2B).
- Further testing to confirm PE may be considered in case of isolated sub-segmental clots (2C).


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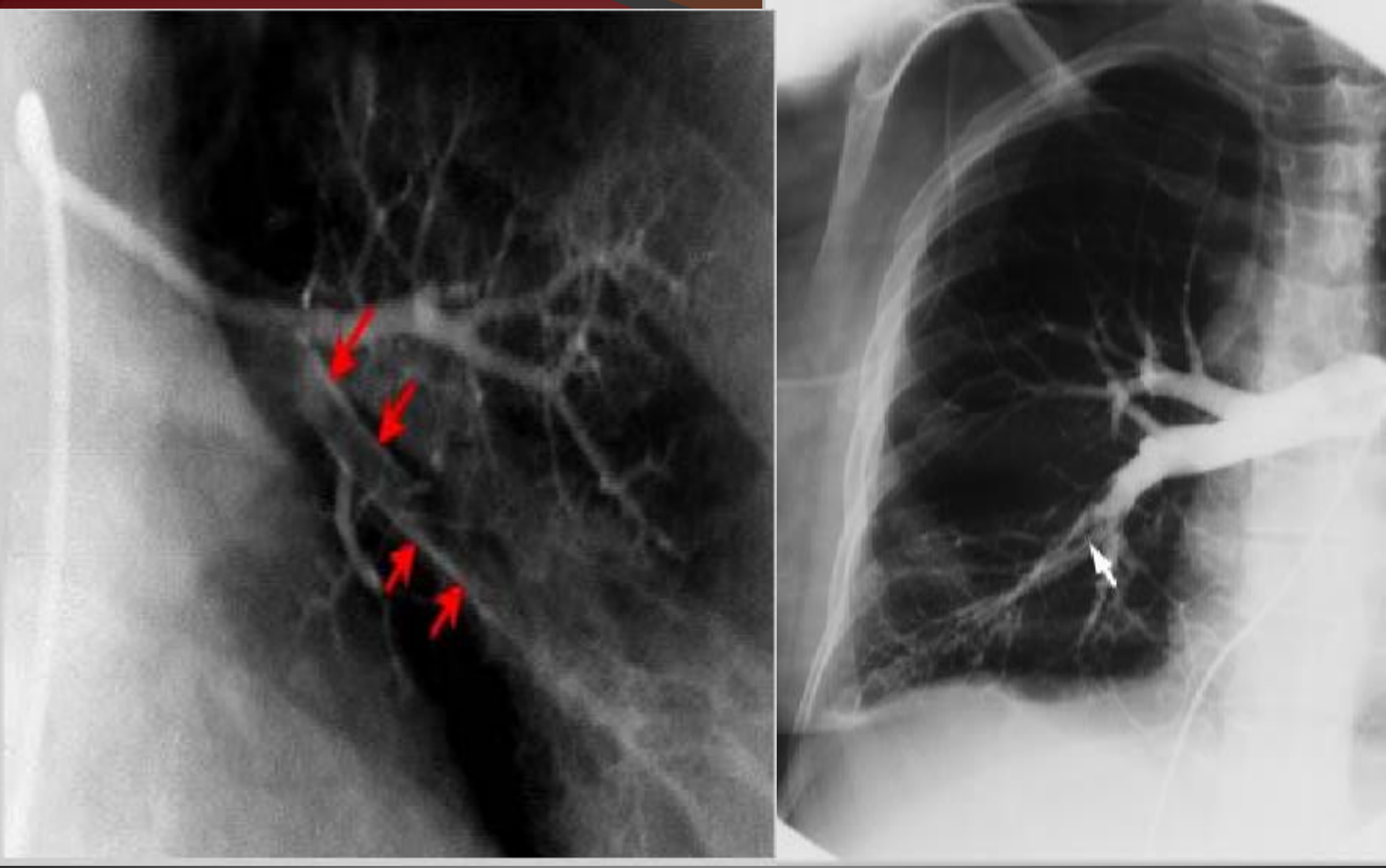


- **The typical appearance of a pulmonary infarct on CT includes a pleural based density with convex borders and a linear strand at the apex of the triangle.**

8- Pulmonary Angiography

- Pulmonary angiography is a reliable but invasive test and is currently useful when the results of non-invasive imaging are equivocal.

- Pulmonary angiography may be considered in cases of discrepancy between clinical evaluation and results of non-invasive imaging tests (2C).



Pulmonary angiogram showing intraluminal filling defects

9- Magnetic Resonance Angiography

- low sensitivity, high proportion of inconclusive MRA scans

- MRA should not be used to rule out PE (3A).

10-Echocardiography:

- In suspected high risk pulmonary embolism

- Hepful in DD of the cause of shock:

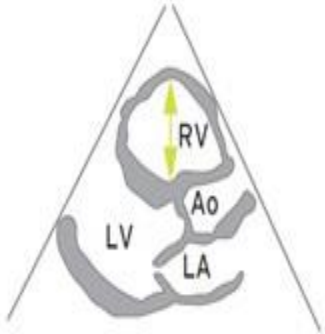
- Pericardial tamponade
- Acute valvular dysfunction
- Severe global or regional LV dysfunction
- Aortic dissection
- Hypovolaemia

- In a patient with shock or hypotension, the absence of echocardiographic signs of RV overload or dysfunction practically excludes PE as a cause of haemodynamic compromise.

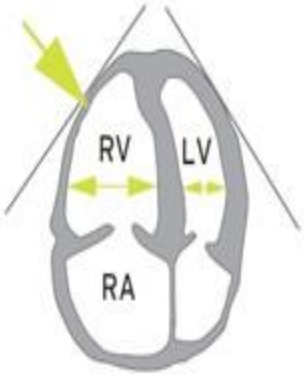
- **Signs of RV overload or dysfunction may also be found in the absence of acute PE and be due to concomitant cardiac or respiratory disease.**

TTE parameters of RV pressure overload (1)

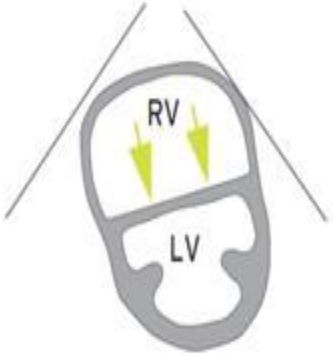
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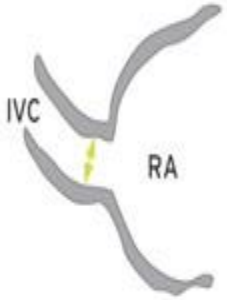
A. Enlarged right ventricle, parasternal long axis view



B. Dilated RV with basal RV/LV ratio >1.0, and McConnell sign (arrow), four chamber view



C. Flattened interventricular septum (arrows) parasternal short axis view

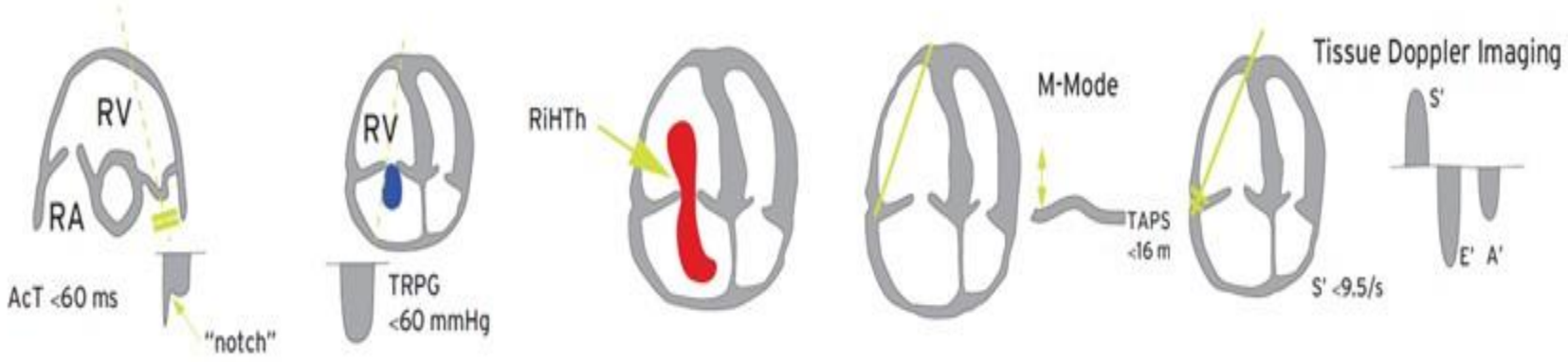


D. Distended inferior vena cava with diminished inspiratory collapsibility, subcostal view

RV = right ventricular; TTE = transthoracic echocardiography/echocardiographic.

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TTE parameters of RV pressure overload (2)



E. 60/60 sign: coexistence of acceleration time of pulmonary ejection <60 ms and midsystolic "notch" with mildy elevated (<60 mmHg) peak systolic gradient at the tricuspic valve

F. Right heart mobile thrombus detected in right heart cavities (arrow)

G. Decreased tricuspid annular plane systolic excursion (TAPSE) measured with M-Mode (<16 mm)

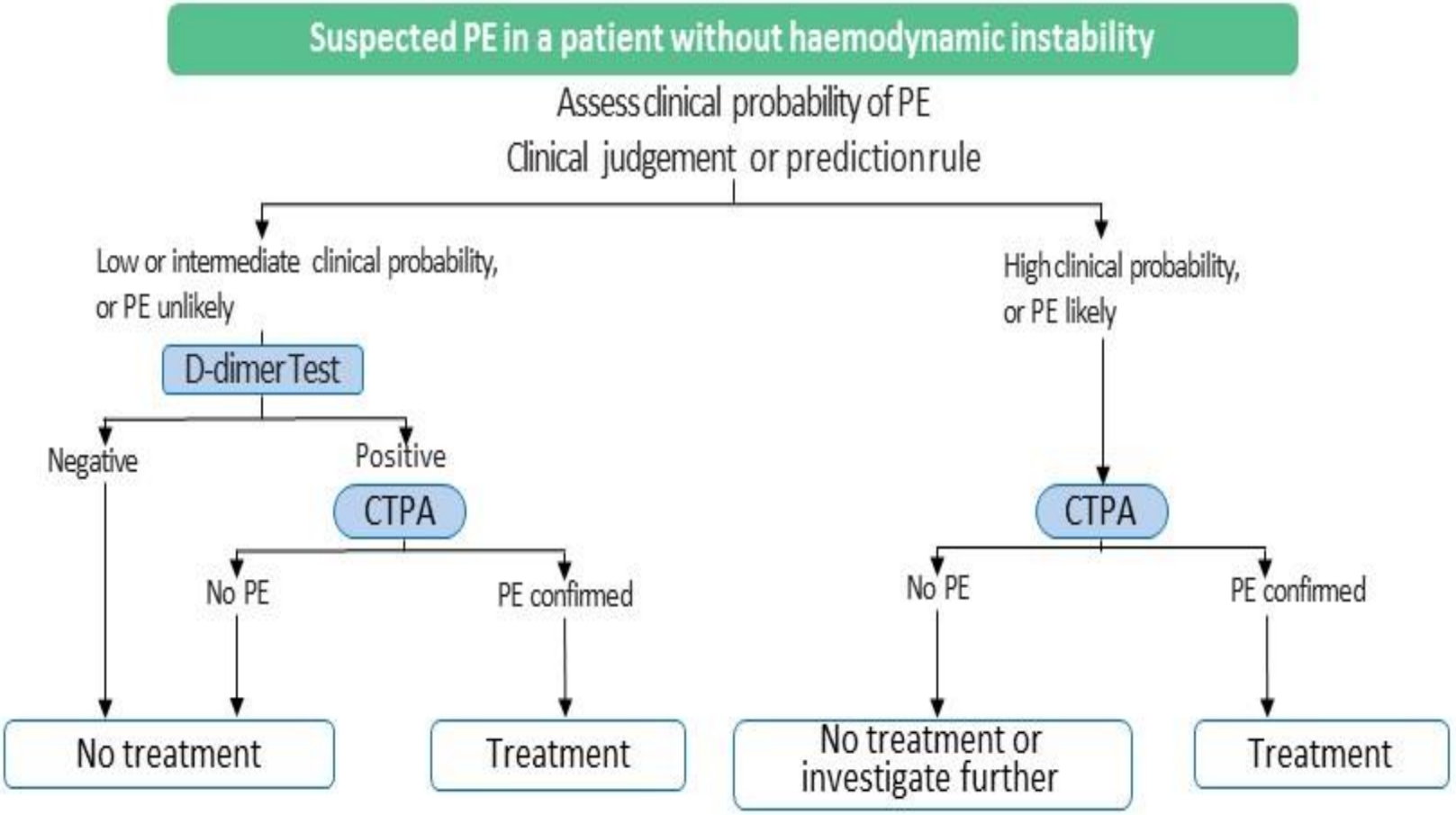
H. Decreased peak systolic (S') velocity of tricuspid annulus (<9.5 cm/s)

RV = right ventricular; TTE = transthoracic echocardiography/echocardiographic.

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Figure 4 Diagnostic algorithm for suspected PE without haemodynamic instability

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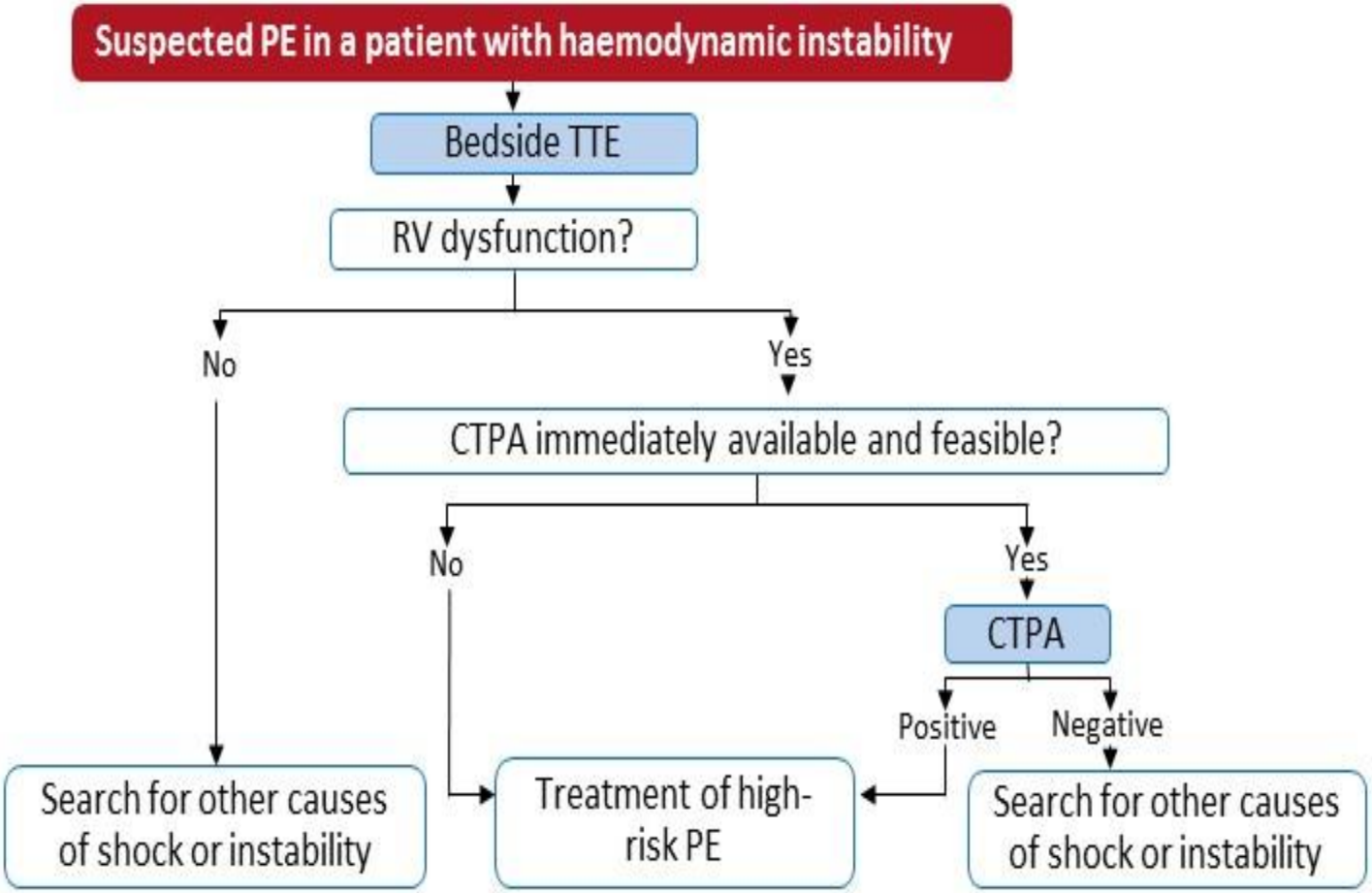
CTPA = computed tomography pulmonary angiography

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Figure 3 Diagnostic algorithm for suspected high-risk PE

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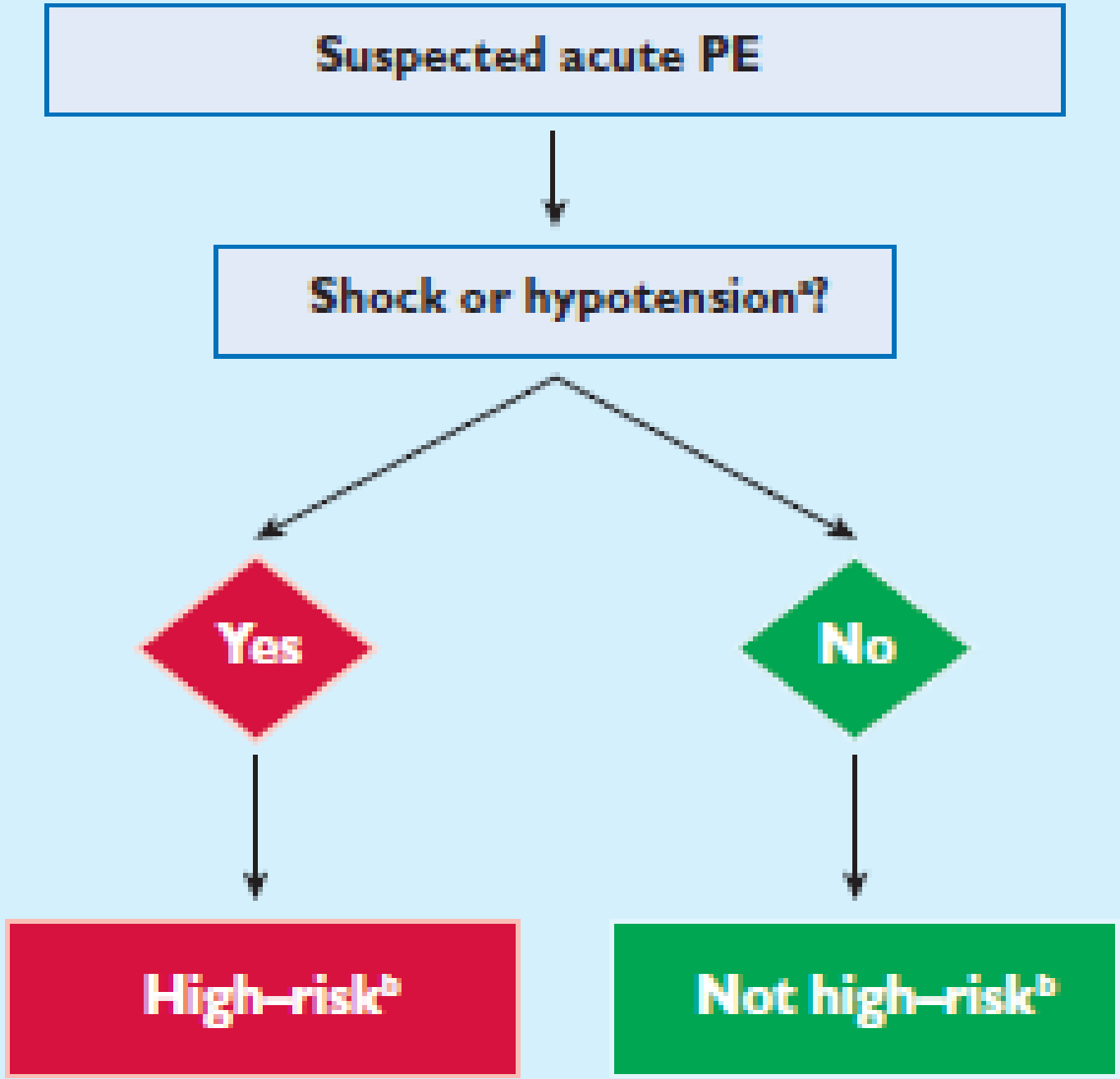
CTPA = computed tomography pulmonary angiography; RV = right ventricular; TTE = transthoracic echocardiography

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Risk Stratification & Prognostic Assessment

- Initial risk stratification of suspected or confirmed PE (based on the presence of shock or persistent hypotension) is recommended to identify patients at high risk of early Mortality (1B).


Suspected acute PE
Shock or hypotension[a]?
Yes
No
High–risk[b]
Not high–risk[b]

Assessment Parameters

A) Clinical parameters:

1. Acute RV failure is a critical determinant of outcome in acute PE
2. Tachycardia,
3. Low systolic BP,
4. Tachypnoea/ Desaturation
5. Syncope

B) Imaging of the right ventricle:

1-Echocardiography:

echocardiographic criteria of RVD as mentioned.

2- Computed tomography:

RV/LV ratio >1.0 increase 30-day PE-related mortality.

3- Laboratory tests & biomarkers

a) Brain natriuretic peptide (BNP):

b) Markers of myocardial injury

- Troponin T and Troponin I: Cardiac troponin elevation was associated higher mortality
- heart type fatty acid binding protein (H-FABP),

Original & Simplified PESI

Parameter	Original version[214]	Simplified version[218]
Age	Age in years	1 point (if age >80 years)
Male sex	+10 points	–
Cancer	+30 points	1 point
Chronic heart failure	+10 points	1 point
Chronic pulmonary disease	+10 points	
Pulse rate ≥110 b.p.m.	+20 points	1 point
Systolic blood pressure <100 mm Hg	+30 points	1 point
Respiratory rate >30 breaths per minute	+20 points	–
Temperature <36 °C	+20 points	–
Altered mental status	+60 points	–
Arterial oxyhaemoglobin saturation <90%	+20 points	1 point

Original and simplified PESI

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	Risk strata	
	Class I: ≤65 points very low 30-day mortality risk (0–1.6%) **Class II: 66–85 points** low mortality risk (1.7–3.5%)	**0 points** = 30-day mortality risk 1.0% (95% CI 0.0–2.1%)
	Class III: 86–105 points moderate mortality risk (3.2–7.1%) **Class IV: 106–125 points** high mortality risk (4.0–11.4%) **Class V: >125 points** very high mortality risk (10.0–24.5%)	**≥1 point(s)** = 30-day mortality risk 10.9% (95% CI 8.5–13.2%)

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PESI = Pulmonary Embolism Severity Index.

Classification of PE based on early mortality risk

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Early mortality risk		Indicators of risk			
		Haemo-dynamic instability	Clinical parameters of PE severity/ comorbidity: PESI III–V or sPESI ≥1	RV dysfunction on TTE or CTPA	Elevated cardiac troponin levels
High		+	(+)	+	(+)
Interme-diate	Intermediate–high	-	+	+	+
	Intermediate–low	-	+	One (or none) positive	
Low		-	-	-	Assessment optional; if assessed, negative

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CTPA = computed tomography pulmonary angiography; PESI = Pulmonary Embolism Severity Index; TTE = transthoracic echocardiography.

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Table 4 Definition of haemodynamic instability

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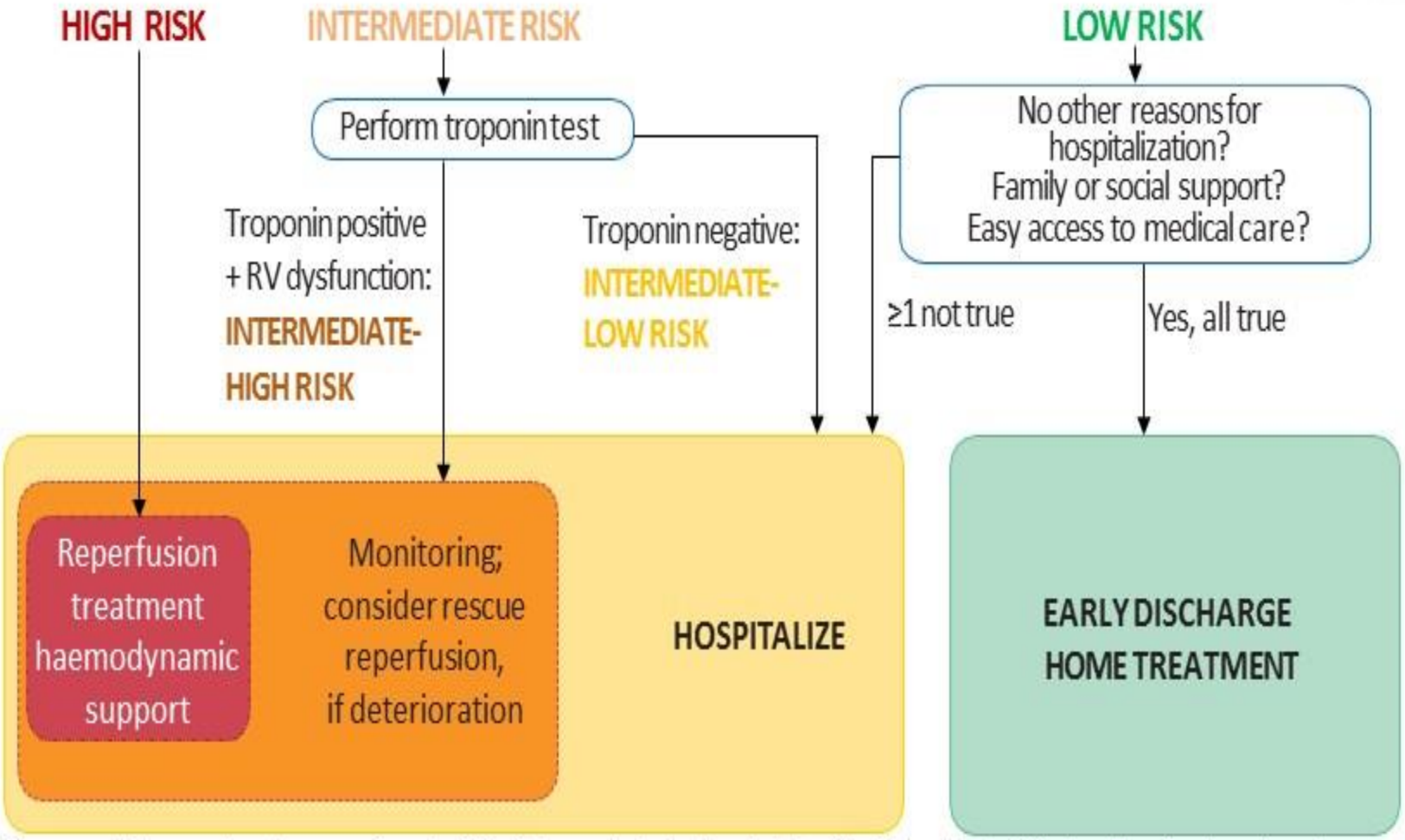
(1) Cardiac arrest	(2) Obstructive shock	(3) Persistent hypotension
Need for cardiopulmonary resuscitation	Systolic BP <90 mmHg, or vasopressors required to achieve a BP ≥90 mmHg despite adequate filling status	Systolic BP <90 mmHg, or systolic BP drop ≥40 mmHg, either lasting longer than 15 minutes and not caused by new-onset arrhythmia, hypovolaemia, or sepsis
	And	
	End-organ hypoperfusion (altered mental status; cold, clammy skin; oliguria/anuria; increased serum lactate)	

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Figure 5 Risk-adjusted management strategy for acute PE (2)

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CTPA = computed tomography pulmonary angiography; PESI = Pulmonary Embolism Severity Index; RV = right ventricular; TTE = transthoracic echocardiography.

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سبحان الله والحمد لله

و لا اله الا الله

والله أكبر و لا حول ولا قوة

الا بالله

Complications of PE:

- Sudden death
- Obstructive shock
- Atrial or ventricular arrhythmias
- Secondary pulmonary arterial hypertension
- Cor pulmonale
- Severe hypoxemia
- Lung infarction
- Pleural effusion

DIFFERENTIAL DIAGNOSIS

- Angina Pectoris
- Myocardial Infarction
- Myocarditis
- Aortic Stenosis
- Atrial Fibrillation
- Cor Pulmonale
- Mitral Stenosis
- Pneumonia
- Pneumothorax
- Acute pleurisy.
- Myositis
- Pleuritis
- Costochondritis
- Rib fracture
- Pericarditis

Treatment

1- Haemodynamic and respiratory support

2- Thrombolytic treatment

3- Anticoagulation:

- Parenteral anticoagulation
- Vitamin K antagonists (VKAs)
- Non-VK oral anticoagulants (NOACs)

4- Percutaneous catheter directed treatment

5- Surgical embolectomy

6- Venous filters

1-Haemodynamic and respiratory support

- Modest fluid challenge (500 ml)
- vasopressors

Inotropic drugs: (norepinephrine, dobutamine, dopamine, epinephrine)

- Vasodilators: inhaled NO, levosimendan.
- Oxygen

2-Thrombolysis

- Thrombolytic treatment of acute PE restores pulmonary perfusion. The haemodynamic benefits of thrombolysis are confined to the first few days
- Thrombolytic therapy is the first-line treatment in patients with high-risk PE presenting with cardiogenic shock and/or persistent arterial hypotension (ESC,ACCP)

What about the submassive?



- Close monitoring is recommended in patients with intermediate-high risk PE to permit early detection of haemodynamic decompensation and timely initiation of 'rescue‘ reperfusion therapy (ESC, IB).

Table 12 Thrombolytic doses and contraindications (1)

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Molecule	Regimen	Contraindications to fibrinolysis
Recombinant tissue-type plasminogen activator (rtPA)	100 mg over 2 h	**Absolute** • History of haemorrhagic stroke or stroke of unknown origin • Ischaemic stroke in previous 6 months • Central nervous system neoplasm • Major trauma, surgery, or head injury in previous 3 weeks • Bleeding diathesis • Active bleeding
	0.6 mg/kg over 15 min (maximum dose 50 mg)	
Streptokinase	250,000 IU as a loading dose over 30 min, followed by 100,000 IU/h over 12–24 h	
	Accelerated regimen: 1.5 million IU over 2 h	

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Table 12 Thrombolytic doses and contraindications (2)

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Molecule	Regimen	Contraindications to fibrinolysis (cont'd)
Urokinase	4400 IU/kg as a loading dose over 10 min, followed by 4400 IU/kg/h over 12–24 h	**Relative** • Transient ischaemic attack in previous 6 months • Oral anticoagulation • Pregnancy or first postpartum week • Non-compressible puncture sites • Traumatic resuscitation • Refractory hypertension (systolic BP >180 mmHg) • Advanced liver disease • Infective endocarditis • Active peptic ulcer
	Accelerated regimen: 3 million IU over 2 h	

BP = blood pressure; IU = international units.

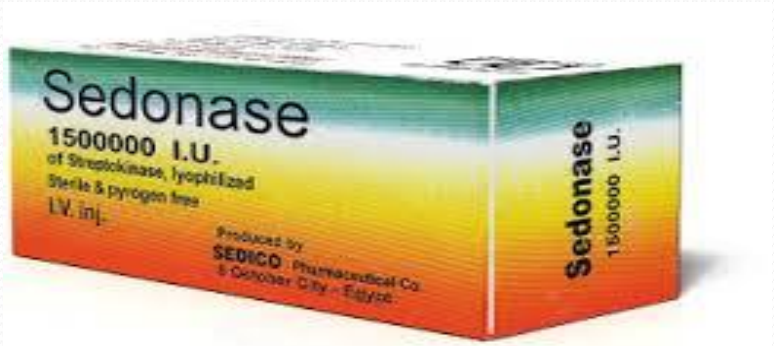
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Contraindications of Thrombolysis (ACCP, 2016)

Major Contraindications[a]
Structural intracranial disease
Previous intracranial hemorrhage
Ischemic stroke within 3 mo
Active bleeding
Recent brain or spinal surgery
Recent head trauma with fracture or brain injury
Bleeding diathesis

Relative contraindications[b]
Systolic BP >180
Diastolic BP >110
Recent bleeding (nonintracranial)
Recent surgery
Recent invasive procedure
Ischemic stroke more than 3 mo previously
Anticoagulated (eg, VKA therapy)
Traumatic cardiopulmonary resuscitation
Pericarditis or pericardial fluid
Diabetic retinopathy
Pregnancy
Age >75 y
Low body weight (eg, <60 kg)
Female
Black race

- Accelerated regimens administered over 2 hours are preferable to prolonged infusions of first-generation thrombolytic agents over 12–24 hours (ESC, ACCP 2C)



- Recombinant tissue plasminogen activators (r-tPAs) include alteplase, reteplase, tenecteplase and desmoteplase, However, alteplase is the only rTPA approved in massive PE (FDA)
- The antidote for tPA in case of toxicity is aminocaproic acid.

- UFH infusion should be stopped during administration of streptokinase or urokinase; it can be continued during rtPA infusion.
- In patients with acute PE who are treated with a thrombolytic agent, we suggest systemic thrombolytic therapy using a peripheral vein over CDT (2C)

3-Anticoagulation

- The standard duration of anticoagulation should cover at least 3 months.

- In some cases, extended anticoagulation beyond the first 3 months, or even indefinitely, may be necessary for secondary prevention, after weighing the individual patient's risk of recurrence vs. bleeding risk.

Parenteral Anticoagulation

- Acute-phase treatment consists of administering parenteral anticoagulation over the first 5–10 days.

1- IV-UFH.

2- S.C. LMWH.

3- S.C. fondaparinux (factor Xa inhibitor).

In patients with high or intermediate clinical probability, immediate anticoagulation should be started

UFH or LMWH/fondaparinux???



- LMWH or fondaparinux are preferred over UFH for initial anticoagulation in PE, as they carry a lower risk of inducing major bleeding and heparin-induced thrombocytopenia (HIT) (ESC,1A) (ACCP, 2B).
- UFH is recommended for:

➢Patients in whom primary reperfusion is considered

➢Serious renal impairment (Cc.Cl < 30 mL/min)

➢Severe obesity.

UFH- ADV

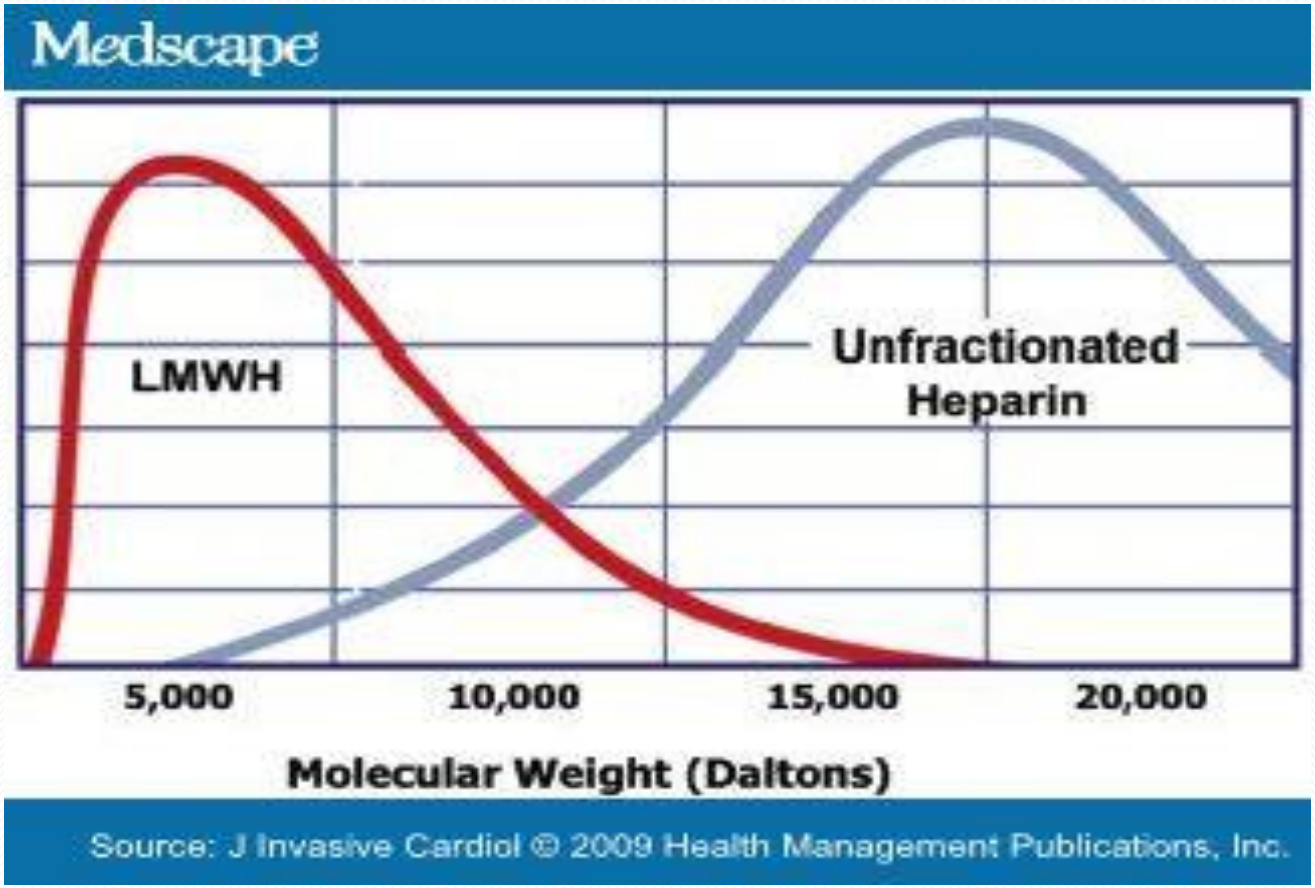
- Short half-life .
- Monitoring its anticoagulant effects.
- Rapid reversal by protamine.

- **Dose:** 80 U/kg (bolus) followed by infusion at the rate of 18 U/kg/h
- The dosing of UFH is adjusted, based on the aPTT
- SC UFH, first dose 333 units/kg, then 250 units/kg without monitoring (2C)

Adjustment of UFH dosage based on the aPTT

Activated partial thromboplastin time	Change of dosage
<35 seconds (<1.2 times control)	80 U/kg bolus, increase infusion rate by 4 U/kg per hour
35–45 seconds (1.2–1.5 times control)	40 U/kg bolus, increase infusion rate by 2 U/kg per hour
46–70 seconds (1.5–2.3 times control)	no change
71–90 seconds (2.3–3.0 times control)	reduce infusion rate by 2 U/kg per hour
>90 seconds (>3.0 times control)	stop infusion for 1 h, then reduce infusion rate by 3 U/kg per hour

- **LMW Heparin: 15,000 kDa (range 3,000-30,000)**
  LMWHs are derived from UFH, using a number of different depolymerization processes (chemical or enzymatic) to cleave the longer saccharide chains.



Methods for Preparation of LMWH

Agent	Method of Preparation
Bemiparin	Alkaline degradation
Dalteparin (Fragmin)	Nitrous acid depolymerization
Danaparoid sodium (Orgaran)	Prepared from animal gut mucosa; contains heparan sulfate (84%), dermatan sulfate (12%), and chondroitin sulfate (4%)
Enoxaparin sodium (Lovenox/Clexane)	Benzylation followed by alkaline depolymerization
Nadroparin calcium (Fraxiparine)	Nitrous acid depolymerization
Tinzaparin (Innohep)	Enzymatic depolymerization with heparinase

Table 11 LMWH and fondaparinux for PE (1)

ESC
European Society of Cardiology

	Dosage	Interval
Enoxaparin	1.0 mg/kg	Every 12 h
	Or	
	1.5 mg/kg	Once daily
Tinzaparin	175 U/kg	Once daily
Dalteparin	100 IU/kg	Every 12 h
	Or	
	200 IU/kg	Once daily

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LMWH = low molecular weight heparin(s).

(European Heart Journal 2019 - doi/10.1093/eurheartj/ehz405)

Table 11 LMWH and fondaparinux for PE (2)

ESC European Society of Cardiology

	Dosage	Interval
Nadroparin	86 IU/kg	Every 12 h
	Or	
	171 IU/kg	Once daily
Fondaparinux	5 mg (body weight <50 kg);	Once daily
	7.5 mg (body weight 50–100 kg);	
	10 mg (body weight >100 kg)	

LMWH = low molecular weight heparin(s).

©ESC

Duration of anticoagulation

Standard (3 months)

Extended

No scheduled stop date

Duration of anticoagulation

- The risk of recurrence if anticoagulants are stopped after 6 or 12 months is similar to that after 3 months
- **Provoked PE:**

VTE provoked by surgery or nonsurgical transient risk factor should receive 3 months of anticoagulant treatment (1B)

- **Unprovoked PE:** in the absence of risk factor.

- **Assessment of the risk of recurrence:**

1) One or more previous episodes of VTE
2) Antiphospholipid antibody syndrome
3) Hereditary thrombophilia
4) Residual thrombosis in the proximal veins.
5) Persistence of RV dysfunction at hospital discharge as assessed by echocardiography

- **A negative D-dimer test one month after withdrawal of VKA seems to be a protective factor for recurrence of VTE**

Indefinite anticoagulant treatment

- **Second episode of unprovoked VTE (IB)**
- **First unprovoked VTE and a low risk of bleeding (IIB)**
- **Patients with cancer (6m LMWH IIB)**
- **Molecular thrombophilia**
  - **Lupus anticoagulant**
  - **Protein C or protein S deficiency**
  - **Homozygous factor V Leiden or homozygous prothrombin G20210A**

- **The option to withdraw anticoagulant treatment should periodically be re-assessed based on the dynamic balance between the risks of recurrence and bleeding (1C)**



- **Lifelong treatment is recommended for most patients with a second unprovoked DVT or PE.**

High bleeding risk

(i) Old age (particularly>75 years),

(ii) Previous GIT bleeding

(iii) Previous stroke (haemorrhagic or ischaemic)

(iv) Chronic renal or hepatic disease

(v) Concomitant antiplatelet therapy (to be avoided, if possible)

(vi) Other serious acute or chronic illness

(vii) Poor anticoagulation control

(viii) Sub-optimal monitoring of anticoagulant therapy

Vitamin K antagonists

- VKAs have been the 'gold standard' in oral anticoagulation & remain the predominant anticoagulants prescribed for PE
- VKAs include:
  - Warfarin (Marevan)
  - Acenocoumarol
  - Phenprocoumon
  - Phenindione (Dendivan)
  - Flunidione

VKA cont.

- Oral anticoagulants should be initiated as soon as possible, and preferably on the same day as the parenteral anticoagulant.
- Anticoagulation with UFH, LMWH, or fondaparinux should be continued for at least 5 days and until the INR has been 2.0–3.0 for two consecutive days.

- Warfarin can be started at a dose of 10 mg in younger healthy outpatients, and at a dose of 5 mg in older patients and in those who are hospitalized.

- The ACCP suggest initiating VKA therapy with warfarin 10 mg daily for the first 2 days followed by dosing based on INR (2C)

- The daily dose is adjusted according to the INR over the next 5–7 days, aiming for an INR level of 2.0–3.0

VKA cont.

Pharmacogenetic testing

- Variations in two genes may account for more than one-third of the dosing variability of warfarin
- One gene determines the activity of cytochrome CYP2C9, that metabolizes warfarin into its inactive form
- The other determines the activity of vitamin K epoxide reductase, produces the active form of vitamin K

- **Paper nomograms or computerized dosing programs (2C, ACCP, 2012)**
- **ACCP recommend against the routine use of pharmacogenetic testing for guiding doses of VKA (1B)**
- **Patients taking VKAs should avoid NSAIDs (including selective COX2) & certain antibiotics (2C)**

Table 3 **Age-adjusted protocol for starting warfarin** [9]

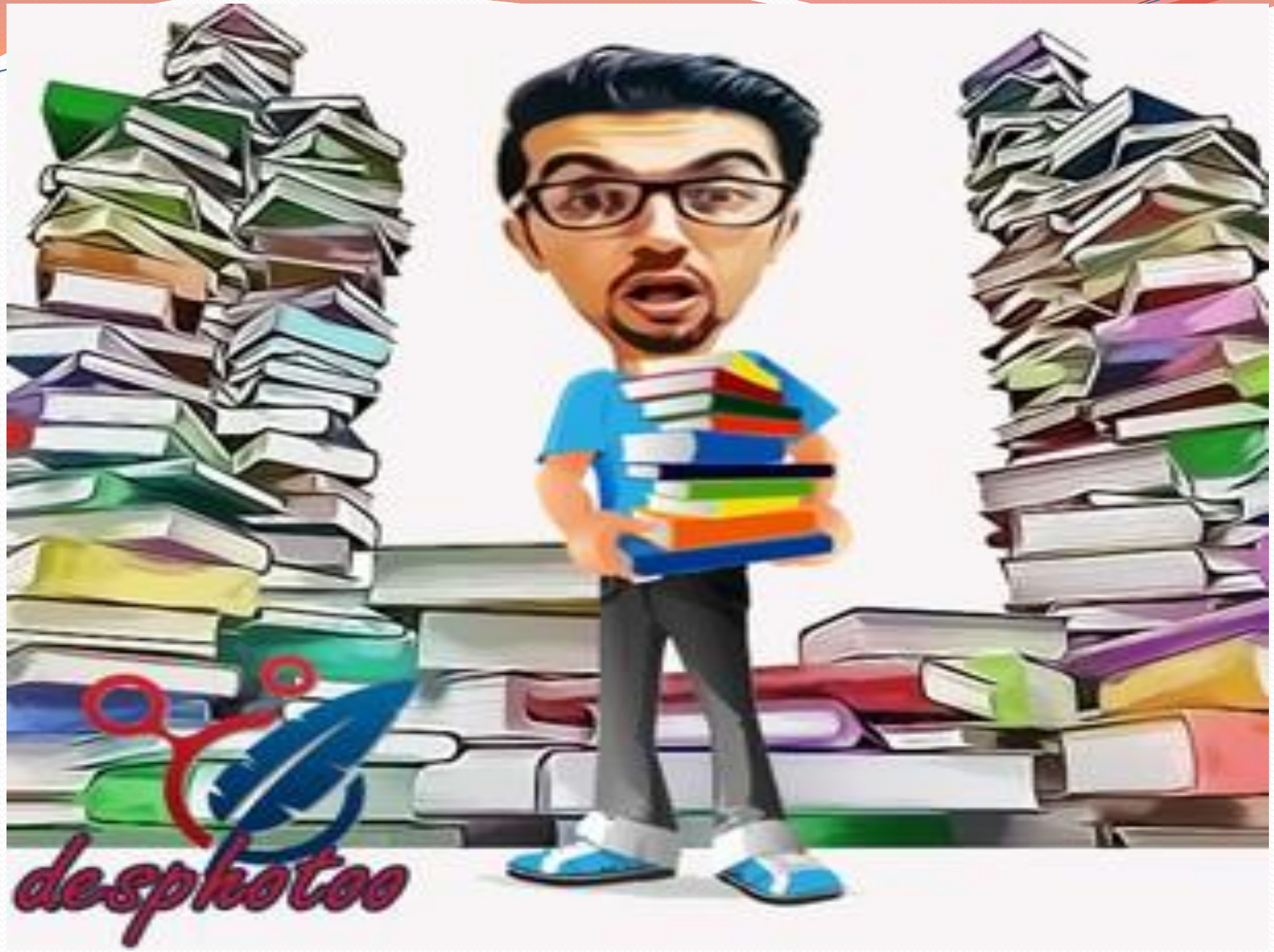
		Dose (mg) according to age			
Day	**INR**	**≤50 years old**	**51–65 years old**	**66–80 years old**	**>80 years old**
1	<1.4	10	9	7.5	6
2	≤1.5	10	9	7.5	6
(16 hours after first dose)	≥1.6	0.5	0.5	0.5	0.5
3	≤1.7	10	9	7.5	6
(16 hours after second dose)	1.8–2.3	5	4.5	4	3
	2.4–2.7	4	3.5	3	2
	2.8–3.1	3	2.5	2	1
	3.2–3.3	2	2	1.5	1
	3.4	1.5	1.5	1	1
	3.5	1	1	1	0.5
	3.6–4.0	0.5	0.5	0.5	0.5
	>4	withhold dose	withhold dose	withhold dose	withhold dose
4	≤1.5	10–15	9–14	7.5–11	6–9
(16 hours after third dose)	1.6	8	7	6	5
	1.7–1.8	7	6	5	4
	1.9	6	5	4.5	3.5
	2.0–2.6	5	4.5	4	3
	2.7–3.0	4	3.5	3	2.5
	3.1–3.5	3.5	3	2.5	2
	3.6–4.0	3	2.5	2	1.5
	4.1–4.5	omit next dose, then	omit next dose, then	omit next dose, then	omit next dose, then
		1–2	0.5–1.5	0.5–1.5	0.5–1
	>4.5	withhold dose	withhold dose	withhold dose	withhold dose

Warfarin dose should be taken at 6 pm each day
Blood for INR should be drawn between 9 and 11 am the next morning

Level of Causation	Anti-infectives	Cardiovascular	Analgesics, Antiinflammatories, and Immunologics	CNS Drugs	GI Drugs and Food
			Potentiation		
Highly probable	Ciprofloxacin Cotrimoxazole Erythromycin Fluconazole Isoniazid Metronidazole Miconazole oral gel Miconazole vaginal suppository Voriconazole	Amiodarone Clofibrate Diltiazem Fenofibrate Propafenone Propranolol Sulfinpyrazone (biphasic with later inhibition)	Phenylbutazone Piroxicam	Alcohol (if concomitant liver disease) Citalopram Entacapone Sertraline	Cimetidine Fish oil Mango Omeprazole
Probable	Amoxicillin/clavulanate Azithromycin Clarithromycin Itraconazole Levofloxacin Ritonavir Tetracycline	Aspirin Fluvastatin Quinidine Ropinirole Simvastatin	Acetaminophen Aspirin Celecoxib Dextropropoxyphene Interferon Tramadol	Disulfiram Chloral hydrate Fluvoxamine Phenytoin (biphasic with later inhibition)	Grapefruit
Possible	Amoxicillin Amoxicillin/tranexamic rinse Chloramphenicol Gatifloxacin Miconazole topical gel Nalidixic acid Norfloxacin Ofloxacin Saquinavir Terbinafine	Amiodarone-induced toxicosis Disopyramide Gemfibrozil Metolazone	Celecoxib Indomethacin Leflunomide Propoxyphene Rofecoxib Sulindac Tolmetin Topical salicylates	Felbamate	Orlistat

Antidote

- **Rapid reversal of anticoagulation with four-factor prothrombin complex concentrate rather than with plasma (ACCP)**
- **Vitamin K 5 to 10 mg administered by slow IV injection**
- IM vitamin K
- The FDA recently approved a plasma-derived product (PCC) for the urgent reversal of VKA toxicity


desphotoo

Non-vitamin K Oral Anticoagulants (NOACs)

(New Oral Anticoagulants)

- Dabigatran
- Edoxaban
- Rivaroxaban (Xarelto)
- Apixaban (Eliquis)

- Dabigatran and Edoxaban are given after acute phase treatment
- Rivaroxaban & apixaban should be started directly (or after a 1–2 day administration of UFH, LMWH or fondaparinux).

NOACs approved for prevention of systemic embolism or stroke in patients with non-valvular AF

	Dabigatran	Apixaban	Edoxaban *	Rivaroxaban
Action	Direct thrombin inhibitor	Activated factor Xa (FXa) inhibitor	Activated factor Xa (FXa) inhibitor	Activated factor Xa (FXa) inhibitor
Dose	150 mg BID 110 mg BID	5 mg BID 2.5 mg BID	60 mg QD 30 mg QD 15 mg QD	20 mg QD 15 mg QD

Pros

- Predictable effect without need for monitoring
- Fewer food and drug interactions
- More predictable half life/elimination
- Improved efficacy/safety ratio

Cons

- The major disadvantage is the lack of specific antagonists in perioperative care and bleeding management
- Lack of Real-World Data for NOAs
- Absolute compliance
- Cost
- Not recommended in patients with severe renal impairment (IIIA).

5- Percutaneous catheter-directed treatment

- Objective: removal of obstructing thrombi from the main pulmonary arteries to facilitate RV recovery and improve symptoms and survival.
- Indications:

➢Absolute contraindications to thrombolysis

➢Failure of thrombolysis

➢intermediate-high-risk patients if the anticipated risk of bleeding under thrombolytic treatment is high (**IIB**).

Catheter interventions without local thrombolysis	
Technique	**Device examples**
Thrombus fragmentation	Pigtail catheter fragmentation Balloon angioplasty using peripheral balloon catheters
Rheolytic thrombectomy	AngioJet 6 F PE® (Bayer, Germany)
Suction embolectomy	Manual aspiration using sheath with detachable haemostatic valve (Argon Medical Devices, Athens, TX, USA)
Rotational thrombectomy	Aspirex® thrombectomy (Straub Medical, Switzerland)
Combined techniques	Pigtail fragmentation (5F) plus AngioJet 6 F PE® thrombectomy (Bayer, Germany)

6- Surgical pulmonary embolectomy

- **Indications:**
  - Contraindications to thrombolysis,
  - Failed thrombolysis
  - Shock that is likely to cause death before thrombolysis can take effect (within hours),

(provided surgical expertise and resources are available) (ACCP, 2C)

7- Venous Filters

- Filters are usually placed in the infrarenal portion of the inferior vena cava (IVC).
- The routine use of IVC filters in patients with PE is not recommended.
- In patients with acute DVT or PE who are treated with anticoagulants, ACCP recommend against the use of an inferior vena cava (IVC) filter (1B)

- **Indications:**

- VTE and absolute contraindication to anticoagulant treatment
- Recurrent PE despite adequate anticoagulation
- Primary prophylaxis in patients with a high risk of VTE.


Thank You